

MRTIS Independent Data-Integrity Audit

Audit white paper — v1, 2026-08-19 02:42 EDT

Subject	MRTIS — Marine/River Traffic Information System (DuckDB warehouse)
Audit type	Independent, read-only, adversarial verification of published claims
Repository state	main @ 06ac44a ("Refine agency fee tiering, extend exclusion list, document the blank-agent cause")
Artifacts audited	scripts/build_db.py , scripts/build_fgis.py , scripts/build_fgis_match.py , scripts/lib/parse.py , sql/schema.sql , sql/schema_fgis_match.sql , dictionaries/*.csv , docs/*.md , CHANGELOG.md
Method	Full clean rebuild in an isolated working copy, then direct SQL/pandas interrogation of the built warehouse against the raw CSV inputs
Isolation	All builds ran in a scratch copy. The project repository, dictionaries and data/db/mrtis.duckdb were not modified. Verified by checksum on every audited file and by git status .

0. Executive summary

Seven claims were tested adversarially — each assumed wrong until the data proved otherwise. **Three are confirmed exactly, one is confirmed with a caveat, and three are wrong.**

The pipeline's mechanical core is sound: it is fully deterministic and idempotent, the FGIS rollup conserves tonnage to the last kilogram, and the "no fuzzy matching" discipline is genuinely absolute — there is no edit-distance code anywhere in the codebase.

The failures are all of one kind: **rules that resolve identity or money by vessel *name* rather than by IMO, and documentation that has drifted ahead of the code.** One corrupted-IMO merge is provably wrong and injects a Baton Rouge workboat's entire 2019 history into a real ocean tanker. The agency-fee rule charges per berth departure rather than per port call, which puts 19.2% of the headline revenue figure in question and contradicts the project's own evidence document. And at least four genuinely commercial vessels sit on the non-commercial exclusion list.

Verdict table

#	Claim	Verdict
1	99.27% of in-coverage FGIS records matched; 1 ambiguous; 0 guesses	Confirmed (one caveat)
2	No fuzzy matching anywhere; <code>DSI Phoenix</code> / <code>DSL Phoenix</code> can never merge	Confirmed
3	31 corrupted-IMO merges are all correct	Wrong — 1 false merge
4	Agency fee correct; no double-counting; no berth sailing mispriced or skipped	Wrong — systematic double-charging
5	Tonnage conserved exactly; every <code>fgis_output</code> line maps to one <code>fgis_record</code>	Confirmed exactly
6	The 28 excluded vessels are all genuinely non-commercial	Wrong — ≥4 real commercial vessels
7	All three scripts idempotent; clean rebuild reproduces every published figure	Split — idempotent yes, figures no

Findings by severity

Sev	ID	Finding	Quantified impact
High	H1	False IMO merge: workboat merged into ocean tanker 9747120	131 rows (51% of all repairs); \$98,000 fabricated fees
High	H2	Agency fee charged per berth departure, not per port call	\$67,259,500 (19.2% of published total); \$1,746,500 indefensible
High	H3	<code>--keep-dredges</code> merges the Texas Star dredge into the real tanker	\$3,937,500 vs \$10,500 correct (375×)
Med	M1	Kennington (IMO 9664926) — real commercial tanker on the exclusion list	583 events, 87 sailings, \$304,500 removed
Med	M2	Name-only exclusions delete real ocean port calls	3 calls (T Jungfrau, Heino, Corinthian)
Med	M3	A missing dictionary silently zeroes revenue, exit code 0	\$349.6M → \$0 or \$313.3M, no warning
Med	M4	<code>DATA_QUALITY.md</code> "Raw rows read" is post-filter, mislabeled	Understates ingest by 23,669 rows
Med	M5	The NULL-fee narrative names six vessels that cannot occur	Hardcoded prose, factually impossible
Low	L1–L10	Dead guard, latent NaN crashes, PK collision path, doc drift	See §8, §9

1. Method

1.1 Environment

The existing `.venv` (Python 3.11.15, duckdb 1.5.5, pandas 3.0.5) was reused. To honour the read-only constraint while still performing the requested full rebuild, the audit ran against an isolated copy:

- `scripts/`, `sql/`, `dictionaries/`, `docs/` and `data/db/mrtis.duckdb` were **copied** into a scratch directory
- The 36 `Zone Report*.csv` files and `fgis_source/` were **symlinked** (read-only inputs)

- All three build scripts were then run inside that copy

Post-audit verification confirmed byte-identical checksums on all six audited code/schema files and all three audited dictionaries, and a clean `git status` for every tracked path.

1.2 Builds performed

Build	Purpose
Baseline (default flags)	Reproduce the published state
Second full rebuild	Idempotency check
<code>--keep-dredges</code>	Behaviour of the documented retention flag
<code>--zone-facility-path</code> missing	Dictionary-absence failure mode
<code>--vessel-type-path</code> missing	Dictionary-absence failure mode

1.3 Verification approach

Each claim was attacked from the data side rather than read from the code: counts recomputed from the raw CSVs, cross-reference integrity tested by independent SQL, merge decisions re-derived from raw row profiles (draft, agent, type, zone, river mile, action mix), and pure functions probed directly with `NaN / None` in every argument position.

2. Claim 1 — Match rate — CONFIRMED

99.27% of in-coverage FGIS records matched; 1 ambiguous; 0 guesses.

Measured

match_status	Records
<code>matched</code>	12,445
<code>unmatched_no_candidate</code>	91
<code>unmatched_ambiguous</code>	1
<code>unmatched_outside_coverage</code>	1,991
Total	14,528

In-coverage = 12,445 + 91 + 1 = **12,537**. 12,445 / 12,537 = **99.2662%** → **99.27%**. Confirmed.

"0 guesses" tested three ways

Test	Result
Records with a <code>vessel_key</code> but <code>match_status <> 'matched'</code>	0
Matched records with NULL <code>vessel_key</code> or NULL <code>mrtis_event_key</code>	0
Unmatched records absent from the review CSV	0 (2,083 = 1,991 + 91 + 1)

The single ambiguous record is `UNMATCHED-AQUITANIA-20241013` — 3 certificate lines, 67,442 MT, corn and wheat to Mexico and Panama. This is precisely the two-real-ships-share-a-name case the design exists to refuse, and it was refused rather than guessed. That is strong corroborating evidence for Claims 2 and 3 as well.

Caveat — identity is never guessed, but event selection is

Vessel identity is never guessed. **Event** selection sometimes is:

Qualifying departures in-window	Matched records
1	11,846
2	462
3	17

For **479 records**, more than one qualifying departure by the correct vessel fell inside the window and `pick_event()` chose by nearest-offset heuristic (ties toward the later date, then lowest `event_key`). A wrong pick attributes tonnage to the wrong sailing — it does not attribute it to the wrong ship. The published claim is defensible as written; this is a precision note, not a defect.

3. Claim 2 — No fuzzy matching — CONFIRMED

No fuzzy matching anywhere. "DSI Phoenix" and "DSL Phoenix" are different real vessels and must never merge.

Exhaustive search for fuzzy machinery

The three scripts, `lib/parse.py` and both SQL schema files were searched for: `levenshtein`, `difflib`, `SequenceMatcher`, `get_close_matches`, `fuzz`, `rapidfuzz`, `thefuzz`, `jellyfish`, `jaro`, `soundex`, `metaphone`, `editdist`, `similarity`, `SQL LIKE / ILIKE / ~ / regexp` on names, and prefix matching via `startswith` / `endswith` on names.

Result: zero hits. The only `startswith` in the codebase is `ship_type_group.startswith("Bulk Carrier")`, on a *ship-type* string, not a vessel name. Every name comparison in the system is exact equality (`==`), set membership (`.isin()`), or a dict key lookup on the normalized string.

Every matching path enumerated

#	Path	Comparison	Fuzzy?
1	FGIS carrier → vessel (resolve)	dict lookup on (date, normalized_name)	No
2	Corrupted-IMO repair (build_imo_repair_map)	exact normalized-name grouping	No
3	Dredge exclusion by name	.isin(set of normalized names)	No
4	Vessel identity fallback (natural_vessel_key)	'NONAME:' + name.strip().upper()	No
5	Alias dedup within a vessel	exact normalized-name equality	No

The specific pair

```
normalize_vessel_name('DSI Phoenix')    -> 'DSIPHOENIX'
normalize_vessel_name('DSL Phoenix')     -> 'DSLPHOENIX'
normalize_vessel_name('D.S.L. Phoenix') -> 'DSLPHOENIX'
```

DSIPHOENIX != DSLPHOENIX under every one of the five paths above. There is no code path in which a one-character difference can bridge them.

In this dataset DSL Phoenix does not appear at all. Dsi Phoenix (IMO 9738337) appears with 20 events and matched 3 FGIS records (2025-12-14, 2026-01-01, 2026-01-02), all correctly and all on the sailing. 21 distinct *PHOENIX* vessels exist in dim_vessel_name_alias; none share a normalized name.

Adjacent finding — str.replace() is not used for editing

The audit brief flagged str.replace() as a risk. It is not used for text editing anywhere. The three .replace() calls are pandas whole-value replacements (Series.replace("", None) , regex=False by default), which substitute the empty-string value and cannot touch substrings.

4. Claim 3 — The 31 corrupted-IMO merges — WRONG (1 of 31)

31 corrupted-IMO merges are all correct. Each merged pair must be one physical ship.

30 of 31 are defensible. One is provably wrong.

4.1 The named guard cases — confirmed correct

Vessel	Outcome	Detail
Aquitania	✓ Separate	9300491 (37 events) and 9611278 (42 events) both retained; both check-digit valid
Sea Voyager	✓ Separate	Four vessels: 9408310 (Tanker), 9799707 (Bulk), 9408762 "Seavoyager" (Tanker), and 8945819 — check-digit invalid and correctly not merged, because two valid twins exist

4.2 H1 — 1782585 → 9747120 ("Egret") is a false merge

This is the largest of the 31 merges: 131 of the 256 repaired rows (51%).

IMO 9747120's own name history in the source data:

Name carried	Rows	Period	Type	Agents
Nord Oceania	28	2020-07-18 → 2025-08-22	Tank	Gulf Harbor, Inchcape, Transmarine
Egret	9	2026-01-01 → 2026-01-09	Tank	Gulf Harbor, Inchcape

The 131 merged rows (raw IMO 17825854 , truncating to 1782585) run **2019-01-03 → 2019-10-21** — before the tanker carried that name, and before the tanker appears in the dataset at all. Their profile is a resident workboat, not an ocean ship:

Signal	Merged 2019 rows	Real tanker 9747120
River miles	226–232 only (6-mile band)	Full river
Enter / Exit (pilot station)	Zero — never enters or leaves the river	Present
Draft recorded	0 of 131 rows	27–42 ft
Agent recorded	0 of 131 rows	Gulf Harbor / Inchcape / Transmarine
Type recorded	0 of 131 rows	Tank
IMO digit distance	7 of 7 digits differ — not a typo pattern	—

Zone distribution of the merged rows: Baton Rouge Gen Anch 51, Apex Baton Rouge 30, Meal Anch 24, Genesis Baton Rouge 10, Placid Oil 8, Dreyfuss 4, Drax Biomass 2, Baton Rouge General Cargo 2.

Consequence in the built warehouse: vessel 9747120 carries 168 events, of which **131 (78%) belong to a different vessel**. 28 of its 32 fee-bearing sailings — **\$98,000 of its \$112,000** — are the workboat's 2019 berth departures, every one of them on a row with no agent recorded.

No FGIS contamination resulted (no FGIS carrier normalizes to EGRET), but the merged workboat did depart Dreyfuss , an Elevator, twice — so it entered the grain-match candidate pool under the tanker's identity.

4.3 Why the guards did not catch it

build_imo_repair_map merges when a normalized name has **exactly one check-digit-valid IMO**. A *second real vessel whose IMO is also corrupt* is invisible to that test — it contributes no valid IMO, so the "exactly one" condition still holds. The three documented guards (one-valid-twin, dredge-name exclusion, type-conflict) do not address this case.

4.4 The other 30 — evidence reviewed

Each merge was profiled on: digit distance and typo shape, overlap between the corrupted rows' dates and the period the good vessel actually carried that name, agent continuity, type continuity, zone overlap, and whether the corrupted rows constitute a complete Enter → berth → Exit river call.

Class	Count	Assessment
Single glitched row inside the good vessel's active window, same agent, same type	24	Correct
Non-overlapping name window, but a complete ocean river call and a 1–2 digit / transposition IMO relationship	6	Defensible
Non-overlapping window, no river call, no draft/agent/type, 7-digit distance	1 (Egret)	Wrong

Strongest positive example: 9139229 → 9139294 ("Happy River") — the raw value is 91392294, which is the correct IMO 9139294 with one duplicated digit; truncation to seven characters then produces the wrong value. Same agent (Norton Lilly), same type (Bulk). Unambiguously one ship.

Suggested discriminator (not implemented — audit is read-only): refuse the merge when the corrupted-IMO rows contain no `Enter / Exit` event **and** their date range does not overlap the period the good vessel carried that name.

5. Claim 4 — Agency fee — WRONG (double-counting)

\$10,500 for bulk, \$3,500 otherwise, only on Depart from a non-anchorage/non-pilot berth. NULL means no fee accrues. Verify no double-counting and no berth sailing wrongly priced or skipped.

5.1 What is correct

The code implements the **stated rule exactly**. No fault was found against the specification as written.

Test	Result
agency_fee non-NULL outside <code>Depart</code> at a berth	0 rows
Non-berth departures (Anchorage / Pilot Station) priced	0 of 56,921
Distinct fee values in the fact table	2 (\$3,500 / \$10,500)
Priced sailings / total	48,167 / \$349,625,500
Zones with NULL <code>facility_type</code> (would be silently skipped)	0 of 215
Rows where the rate disagrees with <code>vessel_type_canonical</code>	0
Under-billing via the ships-register fallback	2 departures — negligible

Nothing is skipped for a missing zone classification, and no row is priced at a rate inconsistent with its vessel type.

5.2 H2 — The fee is charged per berth departure, not per port call

Segmenting `fact_zone_event` on `Enter` (pilot-station) events to delimit port calls:

Charges within one port call	Calls
1	31,025
2	5,667
3	1,144
4	313
5	86
6–10	61
Total fee-bearing calls	38,296

7,271 calls (19.0%) are charged more than once.

Basis	Amount
Booked inside call boundaries (47,899 charges)	\$347,602,500
One charge per call, keeping the highest tier	\$280,343,000
Excess attributable to multi-berth charging	\$67,259,500 — 19.2% of the published total

A further 268 charges (\$2,023,000) precede any `Enter` event in their vessel's history and cannot be attributed to a call at all.

Worked example (vessel_key 287, one `Enter` → `Exit` call, Sep–Oct 2022): Stolthaven 3 → Dow Missouri → Dow Taft , three separate \$3,500 charges between a single river entry and a single river exit.

5.3 The indefensible subset — same berth, minutes apart

179 charges (\$1,746,500) are a second charge at the *same zone* within 60 minutes of the first:

Vessel	Zone	Previous	Charged	Gap	Fee
Bold Guardian	Meraux Buoys Lower	2020-04-27 08:27	08:28	1 min	\$10,500
Fpmc 27	Dow Missouri	2019-06-08 19:28	19:29	1 min	\$3,500
Artvin	ADM Destrehan Buoys Upper	2020-12-15 15:19	15:21	2 min	\$10,500
Mazowsze	175 Buoys	2025-03-06 09:09	09:11	2 min	\$10,500
Oslo Bulk 6	Harmony St	2025-12-05 06:01	06:03	2 min	\$10,500
Santos Eagle	Meraux Buoys Upper	2022-03-27 20:33	20:35	2 min	\$10,500

Widening the window: 186 charges within 6 h (\$1,799,000), 267 within 24 h (\$2,439,500), 586 within 72 h (\$5,271,000).

These survive deduplication because the dedup key (`natural_vessel_key`, `action`, `Time`, `zone_name`) requires an **exact** timestamp match; rows one minute apart are treated as distinct events.

5.4 The project's own evidence contradicts the rule

`docs/PORT_CALL_EVIDENCE.md` line 91, derived from four real Statements of Fact, states the governing principle:

Shifting within a berth is not a second berth call. Ultra Leopard shifted forward and aft at ADM-Reserve several times over the loading (tugs, lines re-made, pilot on/off each time) without ever leaving the berth. Assembly must not read these as separate stops.

The agency-fee logic reads them as exactly that. Whether the fee is contractually per port call or per berth departure is a commercial question the audit cannot settle — but the two documents currently disagree with each other, and \$67.3M sits on the difference.

5.5 M5 — "NULL means no fee accrues" is documented with the wrong vessels

`DATA_QUALITY.md` states the 3 unpriced berth sailings are "*Dixie Raider, Jesse A Mollineaux, Usace Mat Sink Unit, French Warship, Cg Eagle, 'Shop'*". That sentence is hardcoded in `build_db.py` and is now **impossible** — all six are on the exclusion list and cannot reach `fact_zone_event`.

The actual three:

Vessel name	Zone	Facility	Time
Carnival Valor Rb2	Erato St	Cruise	2024-10-19 15:57
Ncl Escape Rb Sb	Julia St	Cruise	2025-12-07 16:36
Ncl Escape Rb Sb	Julia St	Cruise	2025-12-07 17:14

`Carnival Valor Rb2` is the *same physical call* as Carnival Valor (IMO 9236389) at Erato St that day — the real ship arrived 05:35 and departed 16:53. So the semantics claimed ("NULL means *no fee accrues*, not *unknown*") are false for these rows: they are unidentified name-variant artifacts of a call that was already charged. The dollar outcome happens to be correct — charging them would double-count — but for the opposite reason to the one documented.

6. Claim 5 — Tonnage conservation — CONFIRMED EXACTLY

Tonnage is conserved exactly through the FGIS rollup; every `fgis_output` line maps to exactly one `fgis_record`.

6.1 One line, one record

Test	Result
<code>fgis_output</code> rows	18,315
<code>fgis_record_line</code> rows	18,315
Distinct <code>fgis_output_key</code> in the bridge	18,315
Output keys appearing more than once	0
Output keys missing from the bridge	0
Bridge rows referencing a non-existent record	0
Records with no bridge rows	0
<code>sum(fgis_record.line_count)</code>	18,315

Exact bijection. Nothing duplicated, nothing dropped.

6.2 Tonnage

Measure	<code>fgis_output</code>	<code>fgis_record</code>	Delta
Metric tons	544,562,413	544,562,413.0	0.0
Pounds	1,200,553,614,271	1,200,553,614,271.0	0.0
NULL <code>metric_ton</code> rows	0	—	—
NULL <code>pounds</code> rows	0	—	—

Records whose `metric_ton_total` differs from the sum of its own lines (tolerance 1e-6): 0.

Matched tonnage reported in `FGIS_MATCH_QUALITY.md` (473,798,188 MT) reproduces exactly.

6.3 Cross-reference integrity (additional tests)

Test	Result
Matched records whose <code>mrtis_event_key</code> belongs to a different vessel	0
Matched records linked to a non-Elevator/Mid-Stream or non-Depart/Arrive event	0
<code>mrtis_event_action</code> disagreeing with the linked event's action	0
<code>day_offset</code> $\neq$ (event date – cert date)	0
<code>fact_zone_event.fgis_record_count</code> $\neq$ actual record count	0
Events with FGIS records but no stamp	0
Arrive-fallback records that <i>did</i> have a qualifying sailing available	0

The last row independently confirms the fallback discipline — see §9 L4 for the documentation problem attached to it.

7. Claim 6 — The 28 exclusions — WRONG (≥4 commercial vessels)

The 28 excluded vessels are all genuinely non-commercial. Confirm the real Carnival cruise ships and the real tanker Texas Star (9256860) survive.

7.1 The named survivors — confirmed

Vessel	IMO	Events	Fee rows	Fees
Carnival Valor	9236389	1,760	442	\$1,547,000
Carnival Glory	9198367	759	190	\$665,000
Carnival Liberty	9278181	435	108	\$378,000
Carnival Dream	9378474	69	19	\$66,500
Carnival Triumph	9138850	50	13	\$45,500
Carnival Spirit	9188647	12	3	\$10,500
Texas Star (tanker)	9256860	20	3	\$10,500

All survive. The IMO-precise dredge filter works as designed in the default build.

7.2 M1 — **Kennington** (IMO 9664926) is a real commercial tanker

583 events removed at ingest. Its profile is unambiguously commercial:

Signal	Value
IMO	9664926 — check digit valid
Type on every row	Tank
Agent on every row	Celtic (a named shipping agent)
SWP Cross events	174 — roughly 87 river entries/exits, 2019-03 → 2026-07
Primary berth	Crosstex Energy (Tank Storage) — 171 events
Other berths	Phillips 66 Alliance 1
Draft	up to 25 ft
Period	7½ years of continuous shuttle service

A vessel that enters and leaves the river 87 times under a named agent, working a tank-storage terminal, is not a dredge or workboat. It was in the *original* 9-vessel dictionary, so it predates the automated 19-vessel extension.

Impact of removing it: 583 events, 87 berth sailings, \$304,500 in agency fees.

7.3 M2 — Name-only exclusions delete real ocean port calls

The name fallback correctly fires only against rows carrying no valid IMO. But those are precisely the rows on the documented "IMO, Agent and Type go missing together" defective input path — so real vessels land in it.

T Jungfrau — 10 rows, Sep–Oct 2023, a complete loaded tanker call:

```
Enter 09/27/2023 08:12 SWP Cross 25ft
Arrive 09/28/2023 00:37 12 Mile Anch 25ft
Arrive 10/01/2023 08:52 CFI Donaldsonville 106 25ft <- Chemical Plant berth
Depart 10/03/2023 15:59 CFI Donaldsonville 106 42ft <- sailed loaded
Exit 10/04/2023 07:54 SWP Cross 42ft
```

The IMO-bearing **T Jungfrau** (9389289) survives — this is a *different call by the same ship*, deleted because those rows lack an IMO.

Heino — 7 rows, Jun 2020: **Enter** at 25 ft → **Chalmette Slip** (General Cargo) → **Depart** at 21 ft (discharged) → **Exit**. A genuine general-cargo call.

Corinthian — 4 rows, Feb–Mar 2019: **Enter** → **Buck Kreihs** (ship repair) → **Exit**. A real vessel; a shipyard visit is arguably out of scope for cargo analytics, but it is not a dredge.

7.4 The other 24 — correctly excluded

Group	Vessels	Evidence
Dredges / towboats	Mack B (9,733 ev), Allisonk (5,990), Allins K (1,550), Keeneland (709), Randy Martin (502), Dixie Raider (167), Jesse A Mollineaux (99), Sarah Dann, Rayshell Lee, Charles Alfred, Sea Rhino, Vamoose, Capt Ken Scarbrough, Kritical Mass	7–10 ft drafts; minutes-long repeat stops (e.g. Keeneland: 6-minute round trips at Alabo St); no pilot-station transits
Government / naval	Usace Mat Sink Unit, French Warship, Cg Eagle, Rbns Mashhoor (Royal Bahrain Naval Ship)	Non-commercial by nature
Test / non-vessel records	Shop, Abc Test	Obvious junk entries
Tug	Dodge Island (IMO 7917800)	Anchorage / Pilot Station only, 17 ft, zero berth events — no fee impact either way
Cruise tender	Valor Tender Boat 20	Support craft
Dredge with a surviving real twin	Ginny Lab (no IMO) — the IMO-bearing Ginny Lab 8983234 survives with 9 events	Design working as intended

7.5 H3 — **--keep-dredges** merges the Texas Star dredge into the real tanker

The documented retention flag triggers the exact failure the code claims to guard against.

```
$ python3 scripts/build_db.py --keep-dredges
Repaired 32 corrupted IM0(s) affecting 2,645 rows:
...
3110000 -> 9256860      <-- dredge MMSI 311000000, truncated
...
```

Build	Events on 9256860	Chargeable sailings	Agency fees
Default	20	3	\$10,500 
--keep-dredges	2,397	1,125	\$3,937,500  (375x)

Root cause. In `load_dredge_exclusions`, a name is added to the `names` set **only in the `else` branch** — **when the dictionary entry has no usable IMO**:

```
imo = canonical_imo(r.get("imo_raw", "").strip())
if imo:
    imos.add(imo)
else:
    names.add(normalize_vessel_name(name))
```

Texas Star has `imo_raw = 311000000` → canonical `3110000`, so "TEXASSTAR" **never enters** `excluded_names`, and guard #2 in `build_imo_repair_map` never sees it. In the default path this is masked because the dredge rows are filtered out *before* the repair map is built. With `--keep-dredges`, `dredge_exclusions` is `(set(), set())`, so the row filter **and** the name guard disappear together.

As written, guard #2 can never fire for any IMO-bearing dictionary entry.

8. Claim 7 — Idempotency and reproducibility — SPLIT

8.1 Idempotency — CONFIRMED

Two consecutive full rebuilds (`build_db.py` → `build_fgis.py --use-cache` → `build_fgis_match.py`) produced **byte-identical content in all nine tables**:

Table	Rows	Content hash stable
dim_agent	41	✓
dim_vessel	10,211	✓
dim_vessel_name_alias	11,496	✓
dim_zone	215	✓
fact_zone_event	290,436	✓
fgis_output	18,315	✓
fgis_raw	214,213	✓
fgis_record	14,528	✓
fgis_record_line	18,315	✓

The two auto-generated reports and the review CSV regenerate **identical to the committed versions**, modulo the `_Generated ... on <timestamp>_` line:

- docs/DATA_QUALITY.md — identical
- docs/FGIS_MATCH_QUALITY.md — identical
- dictionaries/fgis_match_review.csv — identical (2,083 rows)

8.2 Published figures — NOT REPRODUCED

The auto-generated reports are accurate. The hand-written documents have drifted:

Document	States	Actual
WHITEPAPER §4	9 dredge vessels, ~23,200 rows	28 vessels, 23,669 rows
WHITEPAPER §6	99.2% matched, 94 no-candidate, 99 remaining	99.27%, 91
WHITEPAPER §4	~1–3% of vessels lack a 7-digit IMO	0.4% (39 of 10,211)
WHITEPAPER §1	over 300,000 events	290,436
BUILD.md	246 duplicate rows dropped	230
BUILD.md	70 FGIS records postdate coverage	62
BUILD.md	14.7% multi-certificate	14.8% (1,592 of 10,726)
BUILD.md	9 dredge vessels, ~23,200 rows	28, 23,669
FGIS_MATCH_SPEC	12,442 matched, 94 no-candidate	12,445 / 91
FGIS_MATCH_SPEC	window -1/+4 yields 12,434 matched	12,445
CHANGELOG	48,301 chargeable sailings	48,167
CHANGELOG	window cert_date -1/+3	-1/+4
CHANGELOG	alias table 11,555 rows; 10,270 vessels	11,496; 10,211
CHANGELOG	6,216 of 10,270 vessels matched (60.5%)	6,216 of 10,211 (60.9%)

One apparent discrepancy is **not** an error: WHITEPAPER's "~9% of events have no recorded agent" describes the raw feed (measured: **9.4%**, correct), while DATA_QUALITY.md 's 2.4% describes the built table after dredge removal. Both are right; neither states which population it means.

8.3 M4 — DATA_QUALITY.md "Raw rows read" is mislabeled

The report states **"Raw rows read: 290,666"**. The true count on disk across the 36 files is **314,335** (BUILD.md states this correctly).

stats["rows_in"] is captured *after* the dredge filter runs, so the line beneath it — "Dredge/noise rows filtered out: 23,669" — reads as a deduction from a total that never contained those rows. The arithmetic that actually reconciles is:

$$314,335 \text{ raw} - 23,669 \text{ dredge} - 230 \text{ duplicates} = 290,436 \text{ fact rows}$$

8.4 M3 — A missing dictionary silently zeroes revenue

None of the three dictionary loaders validate; each returns an empty collection and the build completes with exit code 0:

Scenario	Reported agency fees	Warning?
Normal	\$349,625,500	—
zone_facility.csv absent	\$0 ("Chargeable sailings: 0")	None
vessel_type.csv absent	\$313,274,500 (−\$36.4M, −10.4%)	Only "Loaded 0 vessel-type mappings"

Related latent risk: the zone dictionary has 219 rows for exactly 219 distinct zones — 100% coverage **today**. zone_facility.get(z) returns **None** for an unknown zone with no warning, so the first new berth appearing in a future export will silently book \$0 in fees.

9. Low-severity and latent findings

ID	Finding
L1	Guard #2 in <code>build_imo_repair_map</code> is effectively dead. Beyond H3: in the default path the dredge rows are already filtered before the repair map is built, so the excluded names have no IMOs left to protect.
L2	<code>normalize_vessel_name(nan)</code> and <code>canonical_imo(nan)</code> raise <code>AttributeError</code>. Unreachable today — all callers feed <code>.str.strip()</code> output from <code>keep_default_na=False</code> reads. The <code>isinstance</code> -guarded call sites (<code>agency_fee_for</code> , <code>build_imo_repair_map</code>) were probed directly with <code>NaN</code> in every argument position and behave correctly; the two documented NaN bugs do not recur.
L3	<code>fgis_record_id</code> PK collision is reachable on the NONAME path. The schema comment claims uniqueness "by construction" via <code>(imo, cert_date)</code> , which holds only when <code>imo</code> is non-NULL. Two no-IMO vessels differing only in punctuation (<code>ABC-1</code> / <code>ABC 1</code>) have distinct <code>natural_key</code> s but one normalized name; matching both on the same cert date yields a duplicate primary key. Two matched records currently take this path (<code>Strategic Savannah</code> , <code>Sakura L</code>); no collisions exist today.
L4	"The build verifies that no fallback record had a sailing available" is false as stated. <code>BUILD.md</code> , <code>CHANGELOG</code> and the printed <code>FGIS_MATCH_QUALITY.md</code> line ("Verified on every build: zero ") are hardcoded prose. No verification code exists. The property is true and holds structurally — <code>resolve()</code> reaches the fallback only when the departure lookup returns nothing — and the audit confirmed it by independent query, but nothing is checked at build time.
L5	<code>groupby(["name_normalized", "cert_date"])</code> drops <code>NaT</code> keys, after which Phase 2's <code>resolutions[...]</code> lookup would raise <code>KeyError</code> . Not triggered: 0 rows with NULL <code>cert_date_parsed</code> or blank <code>carrier_name</code> .
L6	<code>g_sorted.iloc[-1]</code> selects a vessel's latest name/IMO after <code>sort_values("event_time")</code>, which places <code>NaT</code> last — a single unparseable timestamp would make that row define the vessel's identity. <code>bad_timestamps</code> is 0, so unreachable.
L7	The alias join in <code>load_candidates</code> is time-unbounded — a berth event is indexed under every name the vessel ever carried, regardless of when. This is what makes rename-matching work (11 matched records rely on it legitimately, e.g. <code>ATLANTICISLAND</code> 2019 certs against a name <code>MRTIS</code> first records in 2020), but it is also how a wrong twin could be selected when the right one has a missing berth event.
L8	185 normalized names are shared by more than one <code>vessel_key</code>, 136 of which appear as FGIS carrier names. Date-and-zone narrowing disambiguates all but one (<code>AQUITANIA</code>). Inherent to name-only matching, not a defect — but it is the precision surface.
L9	"Gen" (general cargo, 16,752 rows) maps to Bulk in <code>vessel_type.csv</code> , so general-cargo ships bill at \$10,500 and are reported inside the "Bulk" row of the fee table. Probably intentional given <code>BUILD.md</code> 's General Cargo discussion, but it is stated nowhere.
L10	Dead expression in the fee-table row builder (<code>build_db.py</code>): <code>\${row['count'] and fee['agency_fee'].max() if False else ''}</code> always evaluates to <code>''</code> .

10. Unverifiable from inside this repository

Question	Why it cannot be settled here
Whether <code>Kennington</code> , <code>Dodge Island</code> , <code>Keeneland</code> and <code>Corinthian</code> are commercially agented	A business determination. No external registry covers them — the ships-register snapshot holds only Bulk Carrier and General Cargo rows and all four are absent.
Whether the agency fee is contractually per port call or per berth departure	The audit can show the code double-charges relative to a per-call reading and relative to <code>PORT_CALL_EVIDENCE.md</code> ; it cannot determine which is billed.
Whether the 30 non-Egret merges are each truly one physical ship	The evidence is consistent (typo-shaped IMO, matching name/type/agent, ocean-going behaviour), but external IMO lookup would be needed to settle it.
Whether <code>fgis_source/*.csv</code> matches what USDA currently publishes	<code>--use-cache</code> was used per instruction; no network calls were made.
The 91 no-candidate and 1,991 out-of-coverage FGIS records	Unresolvable without earlier Zone Report exports.

11. Recommended remediation order

Priority	Finding	Rationale
1	H1 — Egret false merge	Corrupts a real vessel's identity, misattributes 131 events and fabricates \$98,000. A targeted guard (no <code>Enter / Exit</code> in the corrupted rows and no overlap with the good vessel's name window) would catch it without disturbing the other 30.
2	H2 — Per-call vs per-berth fee	\$67.3M of the headline figure, and it contradicts the project's own <code>PORT_CALL_EVIDENCE.md</code> . Needs a commercial decision before a code change. The 179 same-berth-within-an-hour charges (\$1,746,500) should be collapsed regardless of that decision.
3	H3 — <code>--keep-dredges</code> guard	One-line fix in <code>load_dredge_exclusions</code> : add the normalized name to <code>names</code> for every excluded entry, not only those without an IMO.
4	M1 / M2 — Exclusion list review	<code>Kennington</code> in particular; then decide whether the name-only rule should spare rows that show a complete <code>Enter → berth → Exit</code> river call.
5	M3 — Dictionary validation	Fail loudly (or warn prominently) when a dictionary is missing or when a zone has no <code>facility_type</code> . A revenue figure should never silently become \$0.
6	M4 / M5 — Report accuracy	Report the true pre-filter row count; derive the NULL-fee narrative from the data instead of hardcoding it.
7	\$8.2 — Documentation drift	Regenerate the hand-written figures from a build, and remove the "the build verifies..." claims or implement the check.

12. Audit integrity statement

- All builds executed in an isolated scratch copy of the repository.
 - `scripts/build_db.py`, `scripts/build_fgis.py`, `scripts/build_fgis_match.py`, `scripts/lib/parse.py`, `sql/schema.sql`, `sql/schema_fgis_match.sql`, `dictionaries/dredge_exclusions.csv`, `dictionaries/vessel_type.csv` and `dictionaries/zone_facility.csv` were verified **byte-identical** before and after the audit.
 - `data/db/mrtis.duckdb` in the project was not rebuilt or modified; its `fgis_record` table still holds the pre-audit 14,528 rows.
 - No file, dictionary or database in the project was changed during the audit. This document and its PDF rendering are the only artifacts added, and were created after the audit concluded at the project owner's request.
 - **Note on repository state:** three files not present at audit start appeared in the repository mid-session — `scripts/build_port_calls.py`, `scripts/lib/guardrails.py`, `sql/schema_port_call.sql` (timestamps 02:35–02:37). They were written by another process, are outside the scope of this audit, and were not examined.
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End of audit white paper — MRTIS `main` @ `06ac44a`, 2026-08-19 02:42 EDT.